

Readability, accuracy and appropriateness and quality of AI chatbot responses as a patient information source on root canal retreatment: A comparative assessment

Mine B  ker ^{*}, Gamze Mercan

Department of Endodontics, Faculty of Dentistry, Mersin University, Mersin, Turkey

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ABSTRACT

Aim: This study aimed to assess the readability, accuracy, appropriateness, and overall quality of responses generated by large language models (LLMs), including ChatGPT-3.5, Microsoft Copilot, and Gemini (Version 2.0 Flash), to frequently asked questions (FAQs) related to root canal retreatment.

Methods: Three LLM chatbots—ChatGPT-3.5, Microsoft Copilot, and Gemini (Version 2.0 Flash)—were assessed based on their responses to 10 patient FAQs. Readability was analyzed using seven indices, including Flesch reading ease score (FRES), Flesch-Kincaid grade level (FKGL), Simple Measure of Gobbledygook (SMOG), gunning FOG (GFOG), Linsear Write (LW), Coleman-Liau (CL), and automated readability index (ARI), and compared against the recommended sixth-grade reading level. Response quality was evaluated using the Global Quality Scale (GQS), while accuracy and appropriateness were rated on a five-point Likert scale by two independent reviewers. Statistical analyses were conducted using one-way ANOVA, Tukey or Games-Howell post-hoc tests for continuous variables. Spearman's correlation test was used to assess associations between categorical variables.

Results: All chatbots generated responses exceeding the recommended readability level, making them suitable for readers at or above the 10th-grade level. No significant difference was found between ChatGPT-3.5 and Microsoft Copilot, while Gemini produced significantly more readable responses ($p < 0.05$). Gemini demonstrated the highest proportion of accurate (80 %) and high-quality responses (80 %) compared to ChatGPT-3.5 and Microsoft Copilot.

Conclusions: None of the chatbots met the recommended readability standards for patient education materials. While Gemini demonstrated better readability, accuracy, and quality, all three models require further optimization to enhance accessibility and reliability in patient communication.

1. Introduction

Persistent periapical pathosis, symptoms in previously treated teeth, or the emergence of periradicular pathology without prior lesions may indicate ongoing endodontic disease [1]. However, this does not necessarily warrant tooth extraction. Comprehensive clinical evaluations, coupled with advanced imaging modalities, can elucidate the underlying etiologies of treatment failure and inform appropriate corrective interventions. In cases where endodontic treatment is deemed inadequate or unsuccessful, root canal retreatment is conducted, which is defined as the procedure involving the removal of root canal filling materials, followed by the cleaning, shaping, and refilling of the canals

[2]. Common causes of failure include the persistence of bacteria in the canals or apex, insufficient mechanical debridement, poor obturation quality, over- or under-extension of the root canal filling, and coronal leakage [3]. These complex procedures require expertise, technical proficiency, and specialized equipment. Referral to an endodontic specialist is often recommended, as it typically provides better long-term outcomes compared to tooth extraction. Providing accurate patient information plays a crucial role in ensuring informed decision-making and improving treatment adherence. COVID-19 pandemic resulted in the widespread adoption of digital communication tools and e-learning and e-assessment methodologies [4]. With the technological advancements brought by the digital age, access to health information sources has

^{*} Corresponding author at: Department of Endodontics, Faculty of Dentistry, Mersin University, Mersin, Turkey.

E-mail addresses: m.buker@hotmail.com (M. B  ker), gmerc7@gmail.com (G. Mercan).

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undergone a profound transformation, with traditional methods being replaced by online platforms, mobile applications, and artificial intelligence(AI)-based tool [5,6].

AI-powered chatbots are software tools that simulate human interaction through text, voice, and visual communication using natural language [7,8], and they have become increasingly popular as tools for providing patient information in healthcare sciences [9]. Particularly in regions with limited access to healthcare services, chatbots stand out as a cost-effective solution. Additionally, they simplify the process for patients to obtain information without directly consulting a healthcare professional, thereby reducing the workload on healthcare systems [10].

The introduction of ChatGPT by OpenAI (San Francisco, CA) in November 2022 marked a significant step toward making artificial intelligence more accessible to the public. This large language model (LLM)-based generative AI tool gained rapid popularity due to its ability to produce detailed and coherent responses. Its success inspired the development of similar tools, including Microsoft's Copilot (Redmond, WA) and Google AI's Gemini (Mountain View, CA). Although these technologies were not initially designed for medical applications, their potential has stimulated widespread interest and experimentation with generative AI methods in healthcare. These chatbots can be integrated to help patients access basic information about their health, learn about their symptoms, and receive quick answers to frequently asked questions [9].

As artificial intelligence becomes increasingly integrated into healthcare and search engines, it is emerging as a significant resource for providing health-related information. AI-based chatbots now compile information from online sources [11], making it more accessible to users seeking health education. While numerous studies have examined the quality and readability of patient education materials derived from websites, the readability of content generated by AI chatbots has not been adequately explored, highlighting a critical gap in current research.

Readability refers to numerical values derived from systematic formulas that indicate the reading level necessary to comprehend written content. Various formulas and tools have been developed to assess readability levels effectively. The readability standards established by organizations such as the American Medical Association, the National Institutes of Health (NIH), and the United States Department of Health and Human Services, patient education materials should ideally be written at or below a sixth-grade reading level [12–14].

Although there are studies evaluating the reliability and validity of chatbot-generated responses in the field of endodontics, no research has been conducted to assess the readability of this information. Our main hypothesis is that the responses generated by the AI chatbots to patients' questions related to root canal retreatment may not meet the recommended readability limits. To test this hypothesis, this study aimed to evaluate quality, accuracy and appropriateness, and readability of AI-based chatbots as a patient information source in root canal retreatment.

2. Materials and methods

2.1. Ethical approval

This study did not require ethical approval as it did not involve human or animal participants.

However, the study adhered to ethical principles, ensuring research integrity and data confidentiality.

2.2. Study design- questions source and generation of Chatbots' answers

This original research study was designed as an observational, cross-sectional study. Ten knowledge-based questions were formulated by two experienced endodontists who routinely perform retreatment procedures, utilizing both their clinical expertise—including frequently encountered patient inquiries in daily practice—and information

derived from the American Association of Endodontists' (AAE) clinical guidelines and educational content on 'root canal retreatment' [15]. Ten frequently asked questions (FAQs) related to root canal retreatment are listed in Table 1.

For the purposes of the study, the three most utilized large language models (LLMs) were selected: ChatGPT-3.5 (OpenAI, San Francisco, CA, US) (<https://chat.openai.com>), Microsoft Copilot (Microsoft, Redmond, WA, US) (<https://copilot.microsoft.com>), and Google Gemini (Version. 2.0 Flash, Google Inc., Mountain View, CA, US) (<https://gemini.google.com/app>). The all-AI applications were accessed via Google's (Google Inc., California, United States of America) search engine query entry.

A single researcher input each question into the chatbots on January 31, 2025, ensuring consistency across all queries. Before the question-and-answer (Q&A) session, all search history and cookies were cleared from the computer. To minimize the effect of previous responses, a new chat window was used for each session. The question set was entered into the systems by a single researcher to ensure consistency. The responses generated by the chatbots were documented immediately following the initial query using Microsoft Office Word 2016 (Microsoft Corporation, Redmond, WA) (Appendix A). All responses obtained by AI chatbots were evaluated for quality, accuracy and appropriateness and readability.

2.3. Readability assessment

Answers were subsequently analyzed for readability using an online calculator. The calculator was accessible via '<https://readabilityformulas.com/free-readability-formula-tests.php>'. We assessed the readability of each response using 7 different indices, including the FRES, FKGL, GFOG, CL, SMOG, ARI, and LW Index [16]. Calculation formulas were presented in Table 2.

The Average Readability Consensus Calculator is a tool used to evaluate the overall readability of a text by combining multiple readability formulas. The Average Readability Consensus Calculator analyses your writing by calculating the number of sentences, words, syllables, and characters. It then applies these metrics to seven popular readability formulas (listed below). These formulas assess the reading difficulty and grade level of your text, helping you determine whether your audience can easily understand your writing.

The Flesch Reading Ease Score (FRES) quantifies text readability by assigning higher scores to content that is more accessible and comprehensible. For example, a score between 90–100 indicates a very easy-to-read text, while a score below 30 suggests an academically challenging text. This formula is widely used by healthcare organizations to create easily readable patient documents [17].

The Flesch-Kincaid Grade Level (FKGL) determines the corresponding U.S. school grade level of the text. For example, a score of 8.0 indicates that the text is suitable for students at the 8th-grade level [18].

The Simple Measure of Gobbledygook (SMOG) estimates the required educational level based on the number of complex words. It is particularly used to assess the readability of healthcare and educational materials [19].

The Gunning Fog Index (GFOG) calculates the grade level by considering the prevalence of long words and complex sentence structures. It is widely used to evaluate the readability of business and technical writing [20].

The Linsear Write (LW) formula determines the readability level based on the ratio of short to long words. It is commonly used to measure the readability of technical documents [21].

The Coleman-Liau Index (CL) estimates the appropriate grade level by analyzing the ratio of letters to sentences within the text. It is used to assess the readability of texts written in digital environments [22].

Lastly, the Automated Readability Index (ARI) calculates the readability level based on the number of letters and words in the text. It can be used to assess the readability of various types of texts [23].

The readability scores for ten responses generated by chatbots were

TABLE 1
List of Frequently Asked Questions.*

1. What is root canal retreatment?
2. In what cases is root canal retreatment required?
3. What are the risks of root canal retreatment?
4. How is the root canal filling material removed?
5. Do I need to use antibiotics before or after retreatment?
6. How many sessions are needed to complete root canal retreatment?
7. Will I experience pain, swelling, abscess,etc. after treatment?
8. What is the success rate of root canal retreatment?
9. What happens if I don't get the root canal retreatment?
10. Is there an alternative treatment to root canal retreatment?

* Questions formulated by two expert endodontists, derived from their daily interactions with patients during routine consultations.

TABLE 2
Readability index formulas used in the study.

Readability index	Formula
Flesch Reading Ease Score (FRES)	$206.835 - 1.015 (\text{total words} / \text{total sentences}) - 84.6 (\text{total syllables} / \text{total words})$
Flesch–Kincaid grade level (FKGL)	$0.39 (\text{total words} / \text{total sentences}) + 11.8 (\text{total syllables} / \text{total words}) - 15.59$
Gunning Fog Index (GFOG)	$0.4 \times [(\text{total words} / \text{total sentences}) + 100 (\text{complex words} / \text{total words})]$
Coleman – Liau Index (CL)	$(0.0588 \times L) - (0.296 \times S) - 15.8$
Simple Measure of Gobbledygook (SMOG)	$1.0430 \sqrt{\text{total polysyllables} (30 / \text{total sentences})} + 3.1291$
Automated readability index (ARI)	$4.71 (\text{characters} / \text{words}) + 0.5 (\text{words} / \text{sentences}) - 21.43$
Linsear Write (LW)	$[(100 - (100 \times \text{nwsy} < 3 / \text{nw})) + (3 \times 100 \times \text{nwsy} >= 3 / \text{total words})] / (100 \times \text{total sentences} / \text{total words})$

L: average number of letters per 100 words, S: average number of sentences per 100 words, nwsy: words 1–2 syllables.

evaluated, with mean (standard deviation) values calculated. These scores were then compared and analyzed against the sixth-grade readability level recommended by the American Medical Association and NIH [12,14]. An accepted readability level was 80.0 for the FRES formula, whereas for the other six formulas, it was set at 6.

2.4. Accuracy and appropriateness and quality assessment

The Global Quality Scale (GQS) was utilized to evaluate overall quality. Evaluations were conducted using a five-point Likert scale, with excellent quality denoted by a score of 5, and low quality by a score of 1, with the following criteria [19]: A GQS score of ‘1’ indicates very poor quality, ‘2’ indicates inferior quality with limited usefulness, ‘3’ is acceptable quality, ‘4’ is high quality, and ‘5’ indicates excellent quality.

To assess the accuracy and appropriateness of responses generated by chatbots, we used the Likert-type scale developed by Tepe and Emekli [24], ranging from 1 to 5 (Table 3). Its development considered multiple critical dimensions: (1) scientific accuracy, ensuring alignment with current scientific knowledge; (2) relevance, evaluating whether the response directly addresses patients’ questions; and (3) actionability, determining if the information provides clear, practical guidance that patients can implement. The accuracy and appropriateness and quality assessment were independently performed by two reviewers. After completing their assessments, they carefully reviewed and discussed any discrepancies in the scoring of answers. Through in-depth discussions, they analysed the specifics of each response, addressing any differences in interpretation. Any disagreements were resolved collaboratively, ensuring a thorough and objective evaluation process. The final score was assigned after reaching a consensus among the evaluators.

2.5. Statistical analysis

Statistical analyses were conducted using Jamovi software (version

Table 3
Likert Scale for Evaluating Chatbot Response Accuracy and Appropriateness.

Score	Accuracy	Description
1	Completely inaccurate	The material contains numerous factual errors, misleading information, or misconceptions that could potentially harm the patient’s understanding or health outcomes.
2	Somewhat inaccurate	While there are some correct elements, the material has significant inaccuracies or omissions that might confuse patients or lead to misunderstandings about their health outcomes.
3	Moderately accurate but lacks clarity or depth	The information is generally accurate, but it lacks sufficient detail on critical points, which could hinder effective self-care or decision-making.
4	Mostly accurate	The material provides accurate information in a clear and understandable manner but may have minor inaccuracies or areas where additional clarification could enhance the patient’s health outcomes.
5	Highly accurate	The information is accurate and well-researched. It comprehensively addresses the topic, enabling patients to fully understand the issue without misconceptions or significant questions remaining.

2.5.28.0). Data were assessed for normality using the Shapiro-Wilk test. For normally distributed continuous variables, one-way ANOVA and Tukey or Games Howell post-hoc test was used to compare means between the groups. Quantitative data was summarized using the mean, standard deviation (SD). Frequency data was shown as number (n) and percentage (%). Sperman’s correlation test was used to assess associations between categorical variables. Statistical significance was set less than 0.05.

3. Results

The readability scores for chatbots’ responses regarding “Root Canal Retreatment” were evaluated using readability calculator. The resulting scores obtained from calculator were statistically analyzed and compared to the recommended 6th-grade reading level.

3.1. Comparison of Chatbots’ readability scores based on seven indices

Table 4 compares the readability levels of texts generated by three different artificial intelligence models: ChatGPT-3.5, Microsoft Copilot, and Gemini. The readability of the texts was measured using seven different readability indices (ARI, FRES, GFOG, FKGL, CL, SMOG, LW).

No statistically significant differences in readability were observed between ChatGPT-3.5 and Microsoft Copilot, as both models produced texts with comparable complexity levels across all seven indices ($p > 0.05$).

There are statistically significant differences between ChatGPT-3.5

TABLE 4

Readability indices for ChatGPT-3.5, Microsoft Copilot and Gemini responses about ‘Root Canal Retreatment’ and statistical comparison of reading scores to 6th grade reading level by using online Calculator (<https://readabilityformulas.com/free-readabilityformula-tests.php>).

Readability Indexes		ChatGPT-3.5	C6thGRL (p)*	Microsoft Copilot	C6thGRL (p)*	Gemini	C6thGRL (p)*	ChatGPT-3.5 vs. Microsoft Copilot (p)†	Gemini vs. Microsoft Copilot (p)†	Gemini vs. ChatGPT-3.5 (p)†
ARI	mean (SD)	14.2 (1.34)	<0.001	14.1 (2.28)	<0.001	10.8 (1.90)	<0.001	0.981	0.002	0.001
FRES	mean (SD)	46.5 (4.81)	<0.001	46.9 (11.93)	<0.001	51.3 (8.26)	<0.001	0.995	0.609	0.282
GFOG	mean (SD)	12.6 (1.06)	<0.001	12.9 (2.31)	<0.001	11.7 (1.54)	<0.001	0.936	0.354	0.257
FKGL	mean (SD)	12.5 (1.22)	<0.001	11.9 (1.99)	<0.001	10.1 (1.54)	<0.001	0.740	0.049	0.009
CL	mean (SD)	12.3 (0.73)	<0.001	12.8 (1.74)	<0.001	11.8 (1.57)	<0.001	0.692	0.387	0.635
SMOG	mean (SD)	11.63 (0.99)	<0.001	11.34 (1.89)	<0.001	9.82 (1.29)	<0.001	0.893	0.064	0.023
LW	mean (SD)	14.0 (1.89)	<0.001	12.8 (2.65)	<0.001	10.4 (2.80)	<0.001	0.536	0.094	0.008
Average reading level score		12.4 (0.70)	< 0.001	12.5 (1.61)	< 0.001	11.0 (1.30)	< 0.001	0.996	0.038	0.046

Bold and italic values statistically significant.

ARI = automated readability index, CL = Coleman-Liau score, FKGL = Flesch-Kincaid grade level, FRES = Flesch reading ease score, FRF = Forcast readability formula, GFOG = gunning FOG, LW = Linsear write, SMOG = Simple Measure of Gobbledygook.

SD: Standard deviation

* C6thGRL(p): Comparison of the responses according to 6th grade reading level (p).

† One-Way ANOVA.

and Gemini in the following readability indices: ARI ($p = 0.001$); FKGL ($p = 0.009$); SMOG ($p = 0.023$); LW ($p = 0.008$). On the other hand, there are no statistically significant differences between ChatGPT-3.5 and Gemini in the following indices: FRES ($p = 0.282$); GFOG ($p = 0.257$); CL ($p = 0.635$).

There were statistically significant differences between Gemini and Microsoft Copilot in the following readability indices: ARI ($p = 0.002$); FKGL ($p = 0.049$). On the other hand, there were no statistically significant differences between Gemini and Microsoft Copilot in the following indices: FRES ($p = 0.609$); GFOG ($p = 0.354$); CL ($p = 0.387$); SMOG ($p = 0.064$); LW ($p = 0.094$).

There was a statistically significant difference in the average reading level (ARL) score, based on seven readability indices, between Gemini and ChatGPT-3.5 ($p = 0.046$) and between Gemini and Microsoft

Copilot ($p = 0.038$), respectively. In contrast, no statistically significant difference was observed between ChatGPT-3.5 and Microsoft Copilot ($p = 0.996$).

3.2. Comparison of Chatbots' answer readability scores to the recommended Sixth-Grade reading level

Based on statistical analysis, there is a statistically significant difference between the readability scores of all three models (ChatGPT-3.5, Microsoft Copilot, and Gemini) and the 6th-grade reading level ($p < 0.001$ for all comparisons across all readability indices). This indicates that the texts generated by all models are significantly more complex than the recommended 6th-grade readability level for patient education materials (Table 4).

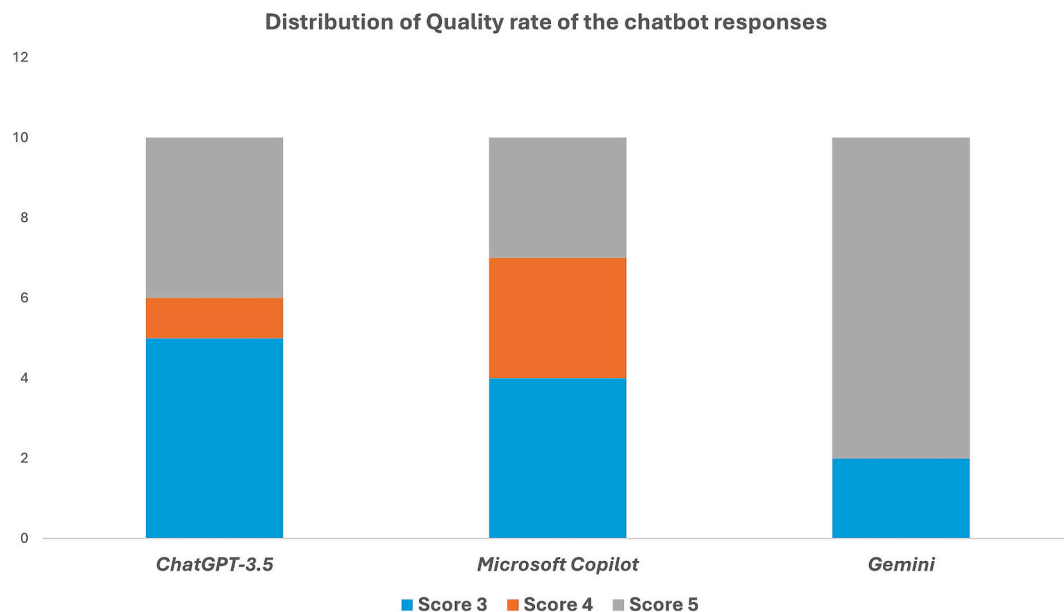


Fig. 1. Distribution of Quality rate of the chatbot responses

3.3. Comparison of quality of chatbots' responses

Based on the responses generated by the LLM chatbots, 40 % (n = 4) of the answers given by ChatGPT-3.5 and 30 % (n = 3) of Microsoft Copilot were excellent quality, while 80 % (n = 8) of Gemini answers were excellent quality (Fig. 1).

3.4. Comparison of accuracy and appropriateness of the chatbots' responses

Based on the responses generated by the LLM chatbots, 20 % (n = 2) of the answers given by ChatGPT-3.5 were correct, while 80 % (n = 8) of Gemini, and 40 % (n = 2) of Microsoft Copilot answers were highly correct. ChatGPT-3.5 and Gemini gave somewhat inaccurate answers to one question each, while Microsoft Copilot gave somewhat inaccurate answers to two question (Fig. 2).

Fig. 3 presents a significant and positive correlation between accuracy and GQS scores ($r = 0.726$, $p < 0.001$).

4. Discussion

The COVID-19 pandemic resulted in significant disparities in oral health, driven by the lack of dental insurance and delays in receiving necessary care. These challenges further deepened existing inequities, underscoring the need for improved access to oral healthcare services. A survey conducted in the United States in 2022 showed that 55 % of adults were dealing with dental problems, and 27 % were unable to visit a dentist due to financial reasons. Among adults who lost their dental insurance, 38 % stated that this was due to job losses or changes related to the COVID-19 pandemic [25]. The suspension of routine dental care to save lives directed patients to seek health-related information through various channels. In this process, patients often used search engines and social media platforms like YouTube to obtain general health information [26,27]. However, in recent years, the use of artificial intelligence chatbots for such inquiries has significantly increased [28]. AI chatbots are valuable tools for providing instant, personalized, and accessible medical information, simplifying complex concepts, and guiding users to appropriate resources, though they are not a substitute for professional medical care.

The findings underscore the challenges associated with the

readability of AI-generated patient education materials in the context of endodontics, particularly concerning root canal retreatment. Our results reveal that the responses provided by ChatGPT-3.5, Microsoft Copilot, and Gemini exceed the recommended sixth-grade reading level for patient education materials, making them less accessible to the general population and supporting the study's hypothesis.

Previous studies have primarily focused on the accuracy or general content quality of AI chatbot responses in endodontics, to the best of our knowledge, the present study is the first to specifically assess the readability of chatbot-generated answers in the context of root canal retreatment. This unique focus highlights an important dimension of AI integration in the development of patient education materials related to in the field of endodontics.

In this study, seven readability indices were used to evaluate the complexity of texts generated by ChatGPT-3.5, Microsoft Copilot, and Gemini. The results showed that all three models produced highly complex texts with low readability scores, exceeding the recommended levels for patient education materials. Gemini had the highest FRES scores compared to other chatbots with no significant differences. This indicates that the texts were comprehensible for 10th to 12th grade students, while their level of difficulty remained accessible to only approximately 54 % of the population [29]. Regarding the Flesch-Kincaid Grade Level, Gemini's responses were found to be appropriate for individuals at 10th grade and above, whereas ChatGPT-3.5 and Microsoft Copilot's responses were suitable for those at 12th grade and above. This suggests that Gemini generated slightly more readable texts compared to both ChatGPT-3.5 and Copilot [30]. The Coleman-Liau Index was similar across all LLM chatbots, indicating that the texts were written at a high school senior level and were difficult to read [22]. SMOG Index values for Microsoft Copilot and ChatGPT-3.5 chatbots showed their responses were at 11th and 12th grade level respectively, and difficult to read for the general majority [31]. In terms of LW and ARI values, Gemini's responses had better readability compared to other models.

The analysis of seven readability indices revealed that all three AI chatbots generated highly complex texts, with readability scores indicating that their content is suitable for high school students or above. Among the models, Gemini provided the most readable responses, with statistically significant improvements in ARI ($p = 0.001$) and FKGL ($p = 0.009$) compared to ChatGPT-3.5, as well as ARI ($p = 0.002$) and FKGL

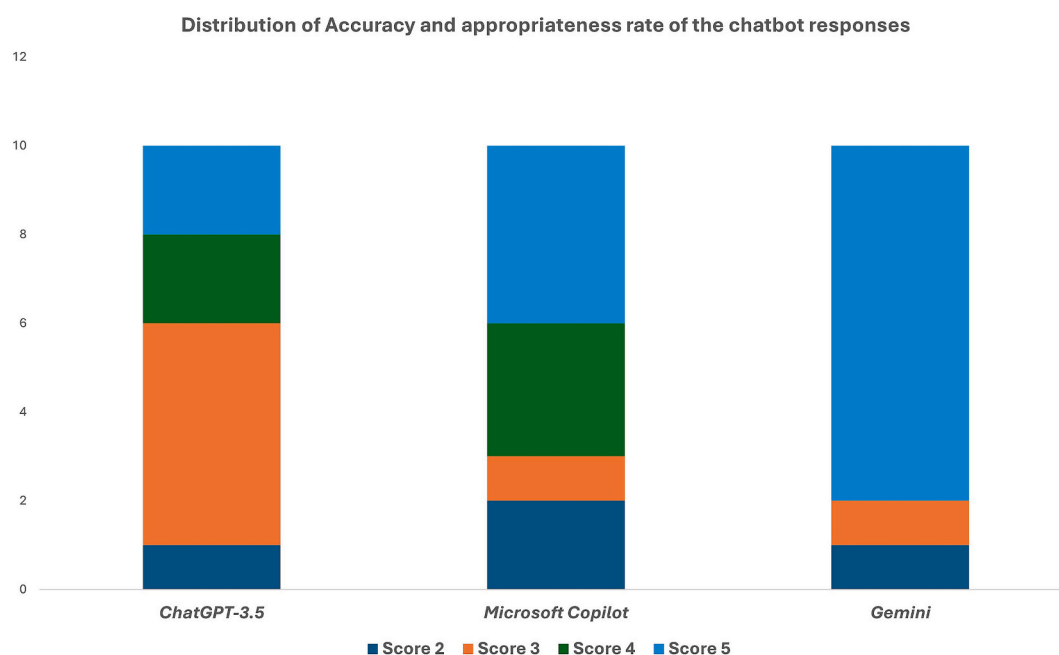


Fig. 2. Distribution of Accuracy and appropriateness rate of the chatbot responses

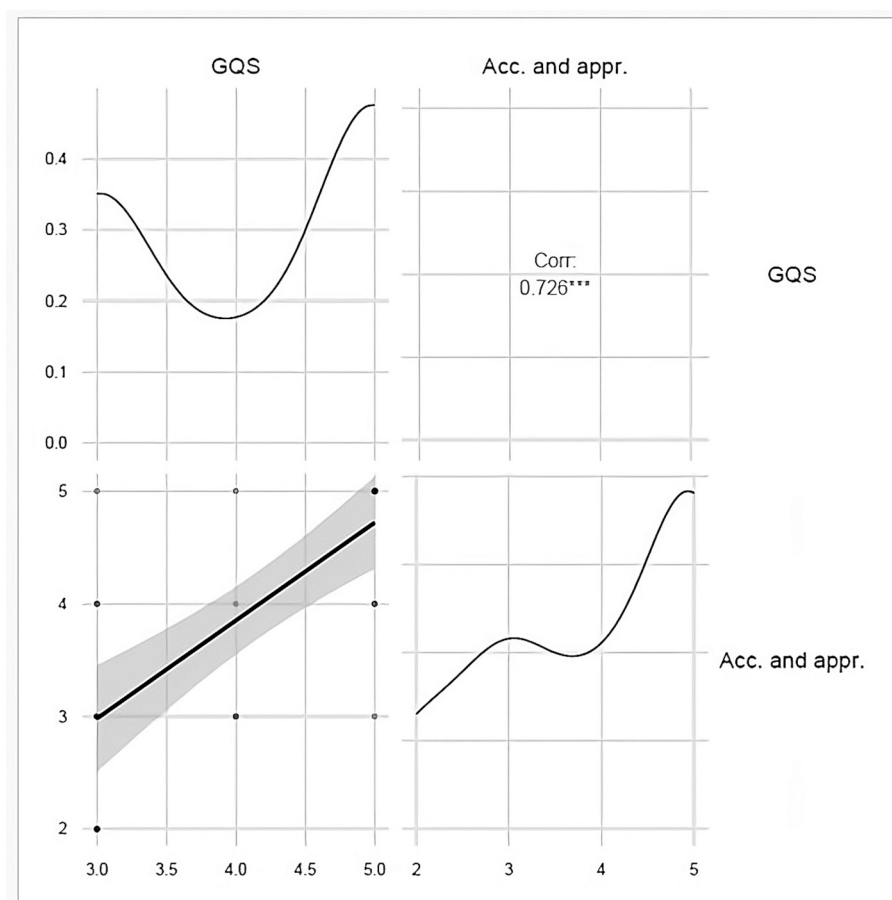


Fig. 3. Correlation of Global Quality Scale score and accuracy and appropriateness score

($p = 0.049$) compared to Microsoft Copilot. Additionally, Gemini demonstrated better readability in SMOG ($p = 0.023$) and LW ($p = 0.008$) indices when compared to ChatGPT-3.5. These results align with findings from previous studies [32–34], which reported that Google Bard (Gemini) provided more comprehensible medical responses than other AI models. This can be attributed to its LaMDA-based architecture, which favors shorter and simpler sentence structures, unlike ChatGPT-3.5, which tends to produce more complex language. Furthermore, no statistically significant difference was detected in the complexity of texts generated by ChatGPT-3.5 and Microsoft Copilot. Several studies in the medical field have also demonstrated that Microsoft Copilot and ChatGPT-3.5 have similar readability scores [35]. These findings are consistent with the results of our study and support the conclusion that both models generate texts of comparable complexity [36]. This similarity suggests that patients can expect comparable reading experiences with both models, potentially due to shared architectural features or training data.

Although Gemini displayed superior readability, all three chatbots generated responses that were challenging for a significant portion of the general population. Our findings are consistent with previous research, indicating that AI-generated medical content generally requires a reading level beyond the recommended threshold for patient education [37–39]. This challenge stems from the fact that AI models are trained on heterogeneous datasets, including academic and professional literature, which often results in complex sentence structures and the inclusion of technical terminology. Previous investigations assessing the readability of AI chatbot responses to patient-related questions across diverse medical and dental fields have indicated that these responses are frequently difficult to understand and generally demand an educational background ranging from high school to college level. The readability of

chatbot responses can be challenging for several reasons. Chatbots are trained on large datasets containing academic, technical, and professional content, which often leads to complex language structures and technical terms in their answers.

While readability is crucial for patient comprehension, accuracy remains a fundamental concern in AI-generated health information. Our analysis revealed that Gemini produced more readable responses, and the accuracy of Gemini's responses was found to be higher compared to the other chatbots. However, the trade-off between readability and accuracy is worth considering [34]. A previous study by Tepe and Emekli [24] found that while Gemini and Microsoft Copilot provided more readable responses regarding breast imaging, ChatGPT 4.0 exhibited greater accuracy. This underscores the importance of verifying chatbot-generated information, as high readability does not always correlate with factual correctness. Additionally, since these tools are continuously updated and modified, the information they generate is fluid and only reflects a specific point in time.

The efficiency of AI-based chatbots may vary across different languages due to differences in linguistic structure, training data coverage, and model optimization. However, there are limited studies investigating the readability of patient education materials in different languages. For example, a study evaluating ChatGPT 4.0's performance in rewriting English and Spanish orthopedic patient education materials found that the AI significantly improved readability by reducing reading grade levels, word count, complex words, and long sentences while maintaining content accuracy [40]. Similarly, a recent study comparing AI-generated Spanish patient education materials with those from the Society of Pediatric Dermatology found that AI tools such as ChatGPT 4.0 and Google Gemini produced content with significantly better readability scores when evaluated at a 6th-grade reading level [41]. In

contrast to the findings of other studies, an investigation in the field of health evaluating Turkish-language responses generated by AI chatbots to biostatistics-related questions found that although the answers tended to be accurate, they were generally difficult to read and understand [42]. This highlights the need for more comprehensive research to evaluate the impact of language in comparing multilingual chatbot performances.

Despite their potential, AI chatbots also have limitations. Errors or inconsistencies in their responses can arise from the operational processes of LLMs [43,44]. For example, ChatGPT's process involves tokenizing input text and using transformer networks to generate responses (<https://openai.com/blog/chatgpt>). Any errors during these steps can result in irrelevant or incorrect answers, which is particularly concerning in critical fields like healthcare. Additionally, chatbots may lack evidence-based sources, often relying on online materials that do not meet rigorous reliability standards. This limitation underscores the importance of verifying the accuracy of chatbot responses before relying on them for medical information.

This study has several limitations. First, only three AI chatbots—ChatGPT-3.5, Microsoft Copilot, and Gemini—were analysed, which may limit the generalizability of our findings. Future research should expand the scope to include additional LLMs and compare their readability and accuracy over time. Second, responses were collected on a single day (January 31, 2025), without accounting for potential model updates that may alter their outputs. Third, while this study focused on readability, comprehensibility was not directly measured using validated tools such as the Patient Education Materials Assessment Tool. Future studies should incorporate such metrics to gain a deeper understanding of how patients interpret and use AI-generated content.

5. Conclusion

AI-based language models represent powerful tools with the potential to enhance patient education; however, the content they generate must be optimized for greater accessibility and accuracy. Gemini produced the highest quality responses, but differences were not statistically significant. Gemini responses were the most reliable, while ChatGPT-3.5 had the lowest reliability. Gemini provided responses with higher readability and accuracy. However further studies with larger datasets would be beneficial to confirm these trends. None of the models fully met the recommended readability standards for patient education materials. Variability in readability and accuracy of responses in the present study suggests that content needs to be aligned with clinical guidelines. Integrating chatbots with current guidelines such as the AAE can increase information reliability. Furthermore, regular evaluations conducted by independent researchers or professional associations, along with the public dissemination of findings, would promote transparency and support patients in making informed decisions.

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Ethics approval statement and document

This study did not require ethical approval as it did not involve human or animal participants

CRedit authorship contribution statement

Mine Bükler: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gamze Mercan:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijmedinf.2025.105948>.

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